# MEKHLOUFI Chaima

Artificial Intelligence & Data Science Engineer (student) | LLM · RAG · Docker · Kubernetes · Linux
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## PROFESSIONAL PROFILE

Final-year AI & Data Science engineering student (ENSIA, Algiers) who builds and ships production-oriented AI systems: on-premise LLM deployments, RAG assistants and containerized microservice platforms on Docker . Solid foundations in Python, NLP, deep learning and reinforcement learning, plus Java/Spring Boot backend experience. Comfortable in both Linux (advanced Ubuntu, shell, server setup) and Windows environments. Open to junior AI / ML / Data / DevOps engineering roles, freelance or part-time, remote or on-site.

## PROFESSIONAL EXPERIENCE

**AI Developer, Incubatee** 2026
*Algeria Telecom Incubator, Algiers*

- Design a privacy-preserving browser extension powered by a local LLM that prevents confidential enterprise data from reaching external AI services.
- Architect an on-device inference pipeline: model quantization, prompt sanitization and context-aware filtering for zero data leakage.
- Evaluate efficient local LLM deployment strategies under strict resource constraints.

**Artificial Intelligence Intern** Sep 2025
*GEAT, Batna, Algeria*

- Designed and deployed a local RAG-based assistant automating Quality Management System (QMS) documentation.
- Built a semantic search engine over internal documents using FAISS and LangChain.
- Integrated and fine-tuned a local LLM adapted to on-premise deployment constraints.

**Data Science & AI Intern** Jun – Jul 2025
*CDTA (Advanced Technologies Development Center), Algiers*

- Processed large-scale 3D point-cloud datasets; applied DBSCAN and K-Means for structure segmentation and classification.
- Developed 3D visualization pipelines that directly supported technical decision-making.

## KEY PROJECTS

**Enterprise Document Management System (DMS)** 2025 – 2026
*Docker · Java / Spring Boot · React · Microservices · REST API · CI/CD · github.com/chaima-m/DMS*

- Designed and deployed a company DMS as containerized microservices (authentication, documents, comments, API gateway, ESB orchestration, translation) with a React/Vite web interface.
- Containerized every service with Docker and Docker Compose; orchestrated on Kubernetes with Deployments, Services, Ingress, ConfigMaps/Secrets and persistent volumes for stateful document storage.
- Applied enterprise-computing practices: autoscaling, high availability (health probes, rolling updates), access control (RBAC) and secure configuration.
- Set up monitoring/logging and automated build-and-deploy workflows with Git and container registries; developed and deployed on Ubuntu/Linux.

**Smart Front-Office RAG Assistant, Algérie Télécom (Forsa Hackathon)** 2025
*Python · FastAPI · ChromaDB · Sentence Transformers · Tesseract OCR · github.com/chaima-m/RAG-ChatBot*

- Built a bilingual (French/Arabic) RAG assistant that answers agents' questions on commercial offers, partner agreements and procedures, with source tracking and responses in under 3 seconds.
- Developed the REST API (FastAPI), OCR ingestion of scanned documents and a web chat interface.

**Arabic Grammar AI Assistant** 2025 – 2026
*Python · Hugging Face · LangChain · Fine-tuning · RAG*

- Curated a domain-specific Arabic grammar corpus and fine-tuned an Arabic LLM for I'rab analysis, sentence correction and rule explanation.
- Built a RAG pipeline for dynamic grammar-rule retrieval, improving factual accuracy.

**E-Commerce Dynamic Pricing with Reinforcement Learning** 2026

*Python · PyTorch · MAML · DQN*

- Implemented a Deep Q-Network for pricing optimization in simulated markets, with MAML for fast adaptation to new product contexts.

**EEG Signal Classification (Brain-Computer Interface)** 2025

*Python · TensorFlow · Keras · MNE*

- Designed CNN and LSTM pipelines for robust classification of EEG signals.

**Timetable Optimization System, ENSIA** 2023 – 2024

*Genetic Algorithms · CSP*

- Built an AI scheduling system for all academic levels, substantially reducing scheduling conflicts.

**Other projects:** K-Means / hierarchical clustering for academic orientation · ARIMA sales forecasting in R · React.js / Tailwind freelance-platform frontend.

## TECHNICAL SKILLS

**AI / ML:** Machine Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, LLM Fine-tuning, RAG

**Frameworks:** PyTorch, TensorFlow, Keras, Hugging Face, LangChain, Scikit-learn, OpenCV, Pandas, NumPy

**Operating Systems:** Linux – Ubuntu (advanced: Bash, server setup, Docker/K8s), Windows (advanced)

**DevOps & Infrastructure:** Docker, Docker Compose, Kubernetes, CI/CD, Git/GitHub, Microservices, Monitoring & Logging, Distributed & High-Performance Computing

**Languages:** Python (advanced), Java, C++, JavaScript,, Bash

**Web & Mobile:** Spring Boot, FastAPI, REST APIs, React.js, Tailwind CSS, Flutter

**Tools:** Jupyter, VS Code, Jira, Asana

## EDUCATION

**Engineering Degree, Artificial Intelligence & Data Science** Sep 2022 – Jun 2027

*ENSIA – National Higher School of Artificial Intelligence, Algiers*

- Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, Big Data & HPC, Distributed Systems, Optimization.
- Winner, Smart Cities Ideathon, Alger ESHRA (Feb 2025): AI-based urban solution.

## LANGUAGES

Arabic: Native | French: Professional (C1) | English: Professional (B2/C1)